

MECHENG726: Acoustics for Engineers

Chapter 4: Acoustic measurement & analysis (ver.1.0)

Learning outcomes

By the end of these lecture notes, students will understand

- ▶ Metrics used for measuring sound level.
- ▶ How to evaluate noise level from measured sound level.
- ▶ How to measure impulse response of an object.
- ▶ Metrics that are used to evaluate acoustical property of rooms.
- ▶ Commonly used measures to calculate reverberation time and speech intelligibility.

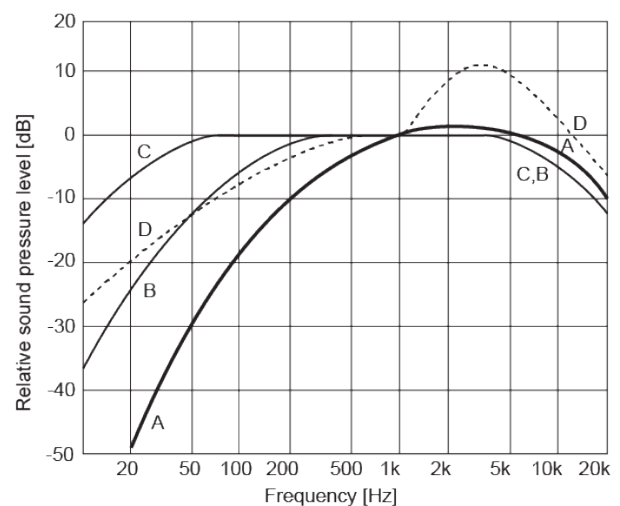
Section 1: Sound level measurement

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Sound level

- ▶ As we learnt in Chapter 2, the amplitude of sound is represented by the sound pressure level (SPL).
- ▶ SPL is a logarithmic measure of sound signal's amplitude in dB relative to the threshold of hearing, i.e. $20 \mu\text{Pa}$.
- ▶ Because subjective volume of sound is frequency-dependent seen by the equal loudness level, the **sound level** is defined by weighting the SPL w.r.t the frequency.
- ▶ Different weighting curves are available: A to D. A is the most commonly used weighting.



Sound level measurement

- ▶ Sound level is measured by processing the signal of recorded sounds (i.e. series of SPL measurements).
- ▶ Various parameters are available for sound level measurement.
 - ▶ **Frequency weighted sound level:** A-weight (L_A), C-weight (L_C), Z-weight (L_Z) (no weight)

Applying a digital filter whose amplitude response follows the corresponding frequency weighting curve to the measured SPL.

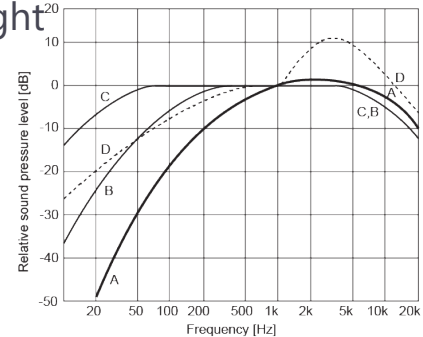
Unit denoted as e.g. **dB(A)** or **dBA** for A-weight

$$L_f(t) = 10 \log_{10} \left(\frac{\mathcal{H}(P(t))^2}{(20 \times 10^{-6})^2} \right)$$

$\mathcal{H}$: Filtering function (e.g. FIR or IIR filters)

$P(t)$: Sound pressure level (SPL) at time instant t

The filter is designed in a way that its frequency response follows the weighting curves.



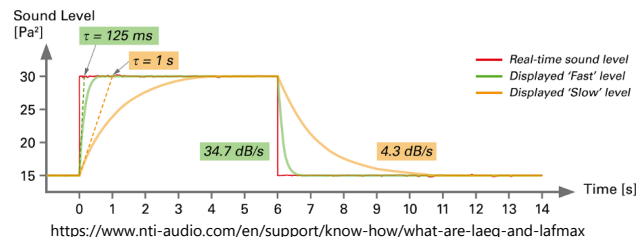
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Sound level measurement

- ▶ **Time weighted sound level:** Fast (L_F), Slow (L_S)
- Used for dampen the sudden change of sound level on display.
Switched by varying the time constant T : Fast (125 ms), Slow (1 s).
Can be applied in conjunction with frequency weight.

$$L_\tau(t) = 10 \log_{10} \left(\frac{\frac{1}{T} \int_{-\infty}^t P^2(\tau) e^{-(t-\tau)/T} d\tau}{(20 \times 10^{-6})^2} \right)$$



<https://www.nti-audio.com/en/support/know-how/what-are-laeq-and-lafmax>

- ▶ There are some other metrics used for sound level

- ▶ **Equivalent continuous sound level:** L_{eq}
- Total sound energy produced over a given period

$$L_{eq}(t) = 10 \log_{10} \left(\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \left(\frac{P(\tau)}{20 \times 10^{-6}} \right)^2 d\tau \right)$$

$P(t)$: Sound pressure level (SPL) at time instant t

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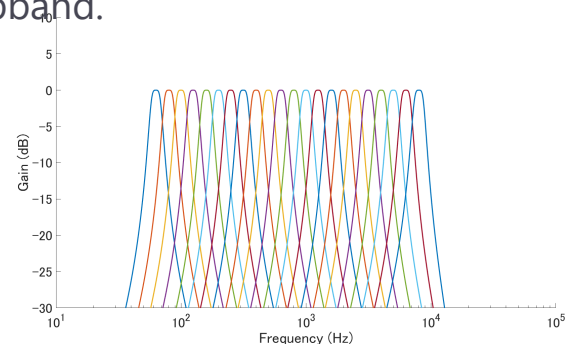
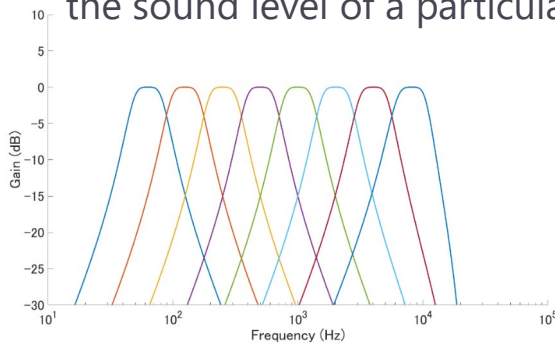
Sound level measurement

- ▶ **Maximum/minimum sound level:** $L_{\max}$, $L_{\min}$
Maximum/minimum value of L_{eq} over time

$$L_{\max} = \max_t L_{\text{eq}}(t)$$

$$L_{\min} = \min_t L_{\text{eq}}(t)$$

- ▶ **Frequency band:** Full-band, octave-band, 1/3 octave-band
For octave- or 1/3 octave-band measurements, corresponding **bandpass filterbank** is applied to the SPL measurement to extract the sound level of a particular subband.



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Room noise criteria

- ▶ **Balanced Noise Criteria (NCB) curve**

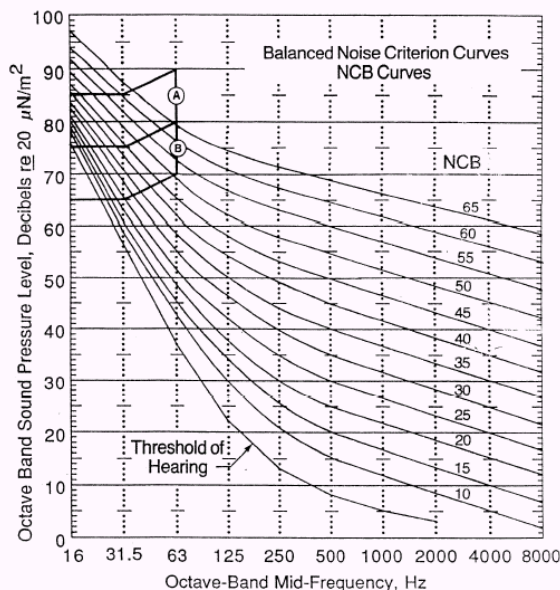


FIG. 2. Balanced noise-criterion (NCB) curves for occupied rooms (new in this paper). Octave-band sound-pressure levels of the magnitudes indicated in regions A and B may induce audible rattle or feeble vibrations in lightweight partitions and ceiling constructions (e.g., thin plaster or gypsum board on metal framing) as follows (taken from Ref. 12): region A: clearly noticeable vibrations and region B: moderately noticeable vibrations.

655 J. Acoust. Soc. Am., Vol. 86, No. 2, August 1989

Leo L. Beranek: Balanced noise criterion (NCB) curves

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TABLE 18.7 Recommended Category Classification and Suggested Noise Criteria Range for Steady Background Noise as Heard in Various Indoor Functional Activity Areas

Type of space (and acoustical requirements)	NC or PNC curve (see Table 18.3 or Table 18.4)	Approximate L_{eq} , dBA
Concert halls, opera houses, and recital halls (for listening to faint musical sounds)	10 to 20	21 to 30
Broadcast and recording studios (distant microphone pickup used)	10 to 20	21 to 30
Large auditoriums, large drama theaters, and large churches (for very good listening conditions)	Not to exceed 20	30
Broadcast, television, and recording studios (close microphone pickup used only)	Not to exceed 25	34
Small auditoriums, small theaters, small churches, music rehearsal rooms, large meeting and conference rooms (for very good listening), or executive offices and conference rooms for 50 people (no amplification)	Not to exceed 35	42
Bedrooms, sleeping quarters, hospitals, residences, apartments, hotels, motels, etc. (for sleeping, resting, relaxing)	25 to 40	34 to 47
Private or semiprivate offices, small conference rooms, law offices, libraries, etc. (for good listening conditions)	30 to 40	38 to 47
Living rooms and drawing rooms in dwellings (for conversing or listening to radio and television)	30 to 40	38 to 47
Loft offices, reception areas, retail shops and stores, cafeterias, restaurants, etc. (for moderately good listening conditions)	35 to 45	42 to 52
Lobbies, laboratory work spaces, drafting and engineering rooms, general secretarial areas (for fair listening conditions)	40 to 50	47 to 56
Light maintenance shops, office and computer equipment rooms, kitchens and laundries (for moderately fair listening conditions)	45 to 55	52 to 61
Shops, garages, power plant control rooms, etc. (for just acceptable speech and telephone communication). Levels above NC- or PNC-60 are not recommended for any office or communication situation	50 to 60	56 to 66
For work spaces where speech or telephone communication is not required, but where there must be no risk of hearing damage	60 to 75	66 to 80

Beranek, Leo L. "Balanced noise-criterion (NCB) curves." *The Journal of the Acoustical Society of America* 86.2 (1989): 650-664.

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Practical example

- ▶ *Sound level meter* is normally used for measuring sound level.
- ▶ It is a device with a high precision omni-directional microphone attached to the main body equipped with a display and processor.
- ▶ Most sound level meter offers measurements with various parameters we have discussed.

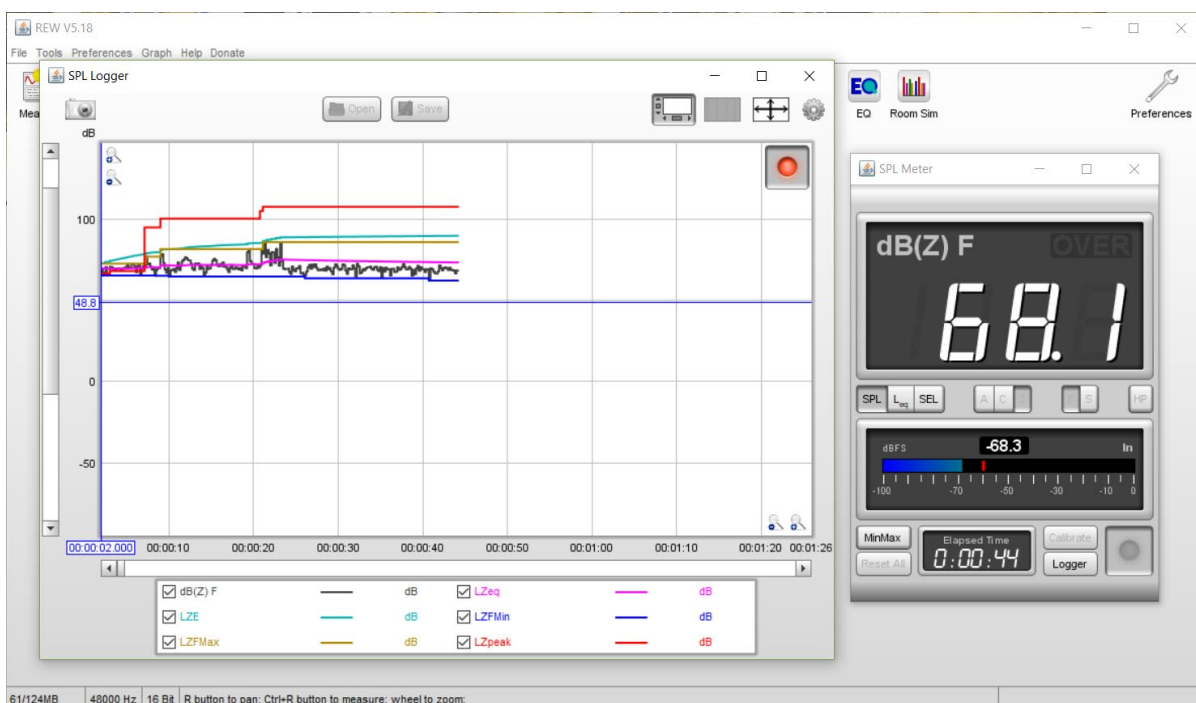


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Practical example

- ▶ Sound level meter in REW



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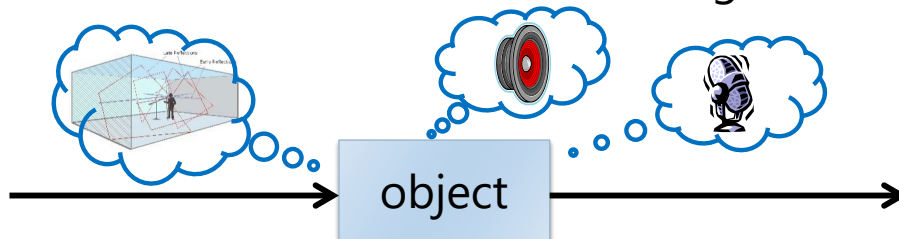
Section 2: Impulse response measurement

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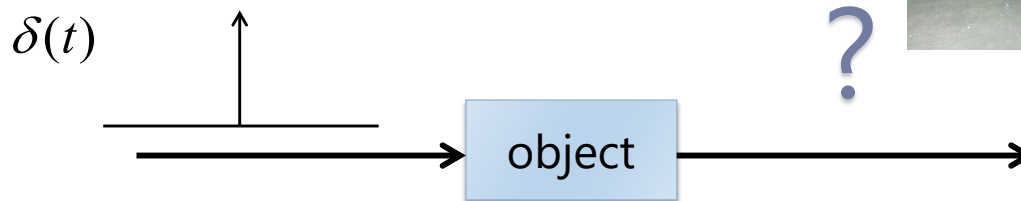
Measuring impulse responses

- ▶ Actual behaviour of the objects in acoustics (microphones, rooms, etc.) are usually unknown until their properties are practically measured.
- ▶ Since the impulse response and corresponding frequency response describe the behaviour of an object, **measuring the impulse response** will enable us to analyse properties of the object quantitatively as well as objectively.
- ▶ Impulse response measurement is a fundamental skill that we need to know as an acoustics engineer.



Impulse response measurement: how?

- ▶ The simplest way to measure the impulse response of an object is literally measuring the “response to an impulse”, which can be realised by exciting the object by a unit impulse.
- ▶ Unit impulse can be projected as **sound wave**, **electrical current**, or **force** depending on the type of the object.
 - ▶ Sound wave: microphones, rooms
 - ▶ Electrical current: loudspeakers, amplifier circuits
 - ▶ Force: solid material

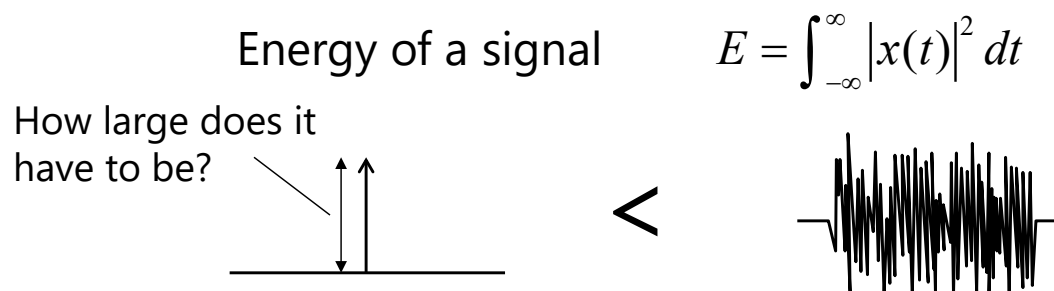


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Impulse response measurement: how?

- ▶ Using unit impulse for impulse response measurement is less common because of the effect of **noise**, which cannot be avoided in practical acoustical measurements.
- ▶ Since the energy of a unit impulse can reside only in the specific time instant (i.e. $t=0$), it is impossible to increase the energy of the signal, which reduces the signal-to-noise ratio (SNR).



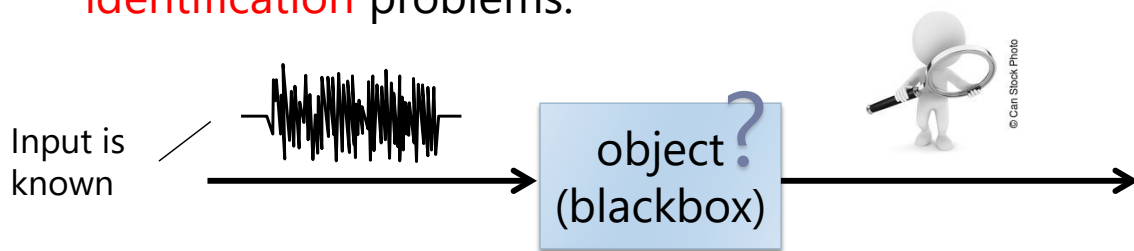
Amplitude of unit impulse has to be very large to achieve better SNR.

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Impulse response measurement: how?

- ▶ In order to overcome the noise issue, various techniques have been invented. Some of them are commonly used in practical acoustic measurements.
- ▶ Although each technique uses different input signals and processes, the basic principle behind them are the same.
- ▶ The idea is “estimate the behaviour of a blackbox (object) by observing its output provided the object is excited by a known input”. This is also known as a part of **system identification** problems.

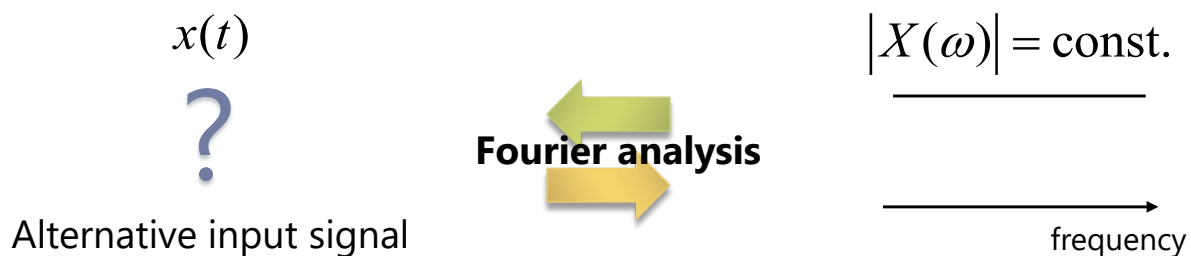


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Requirement for the input signal

- ▶ The input signal has to replace the unit impulse while overcoming the issue with low SNR.
- ▶ In theory amplitude spectrum of unit impulse is constant for all ω . The input signal also has to have a uniformly distributed amplitude spectrum.



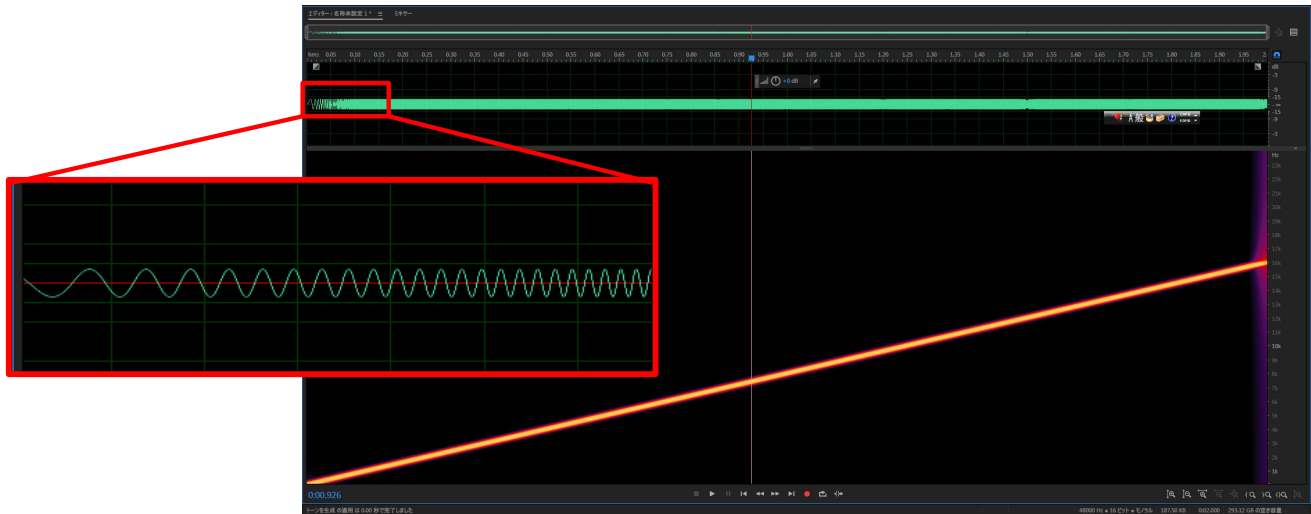
- ▶ A few signals satisfying this requirement are used as the input signal for the impulse response measurement.

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Swept sine signal

- ▶ **Swept sine** is a sinusoidal signal the frequency of which varies with respect to the time elapsed.
- ▶ It is commonly used for impulse response measurements, and is also known as **chirp signal** or **time stretch pulse**.

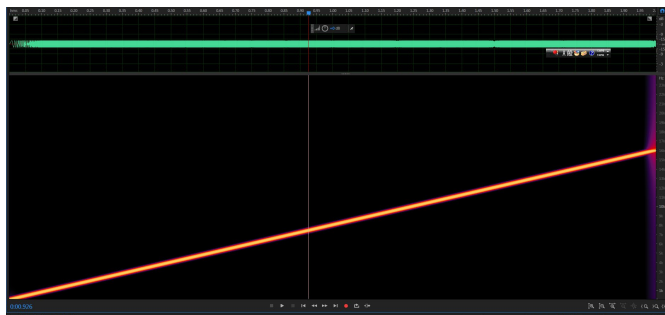


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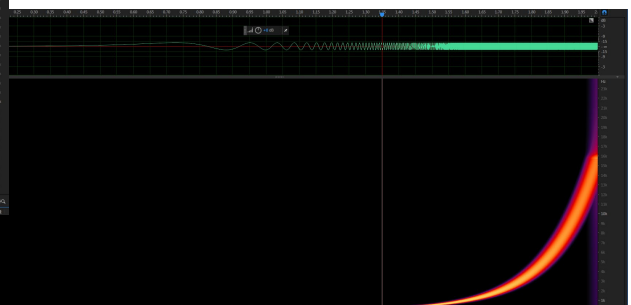
Swept sine signal

- ▶ A few different types are available depending on how the frequency varies.
 - ▶ Linear swept sine
The frequency varies **linearly** with respect to time elapsed
 - ▶ Log swept sine
Frequency varies **logarithmically** with respect to time elapsed



Linear swept sine (2 sec)

Log swept sine (2 sec)

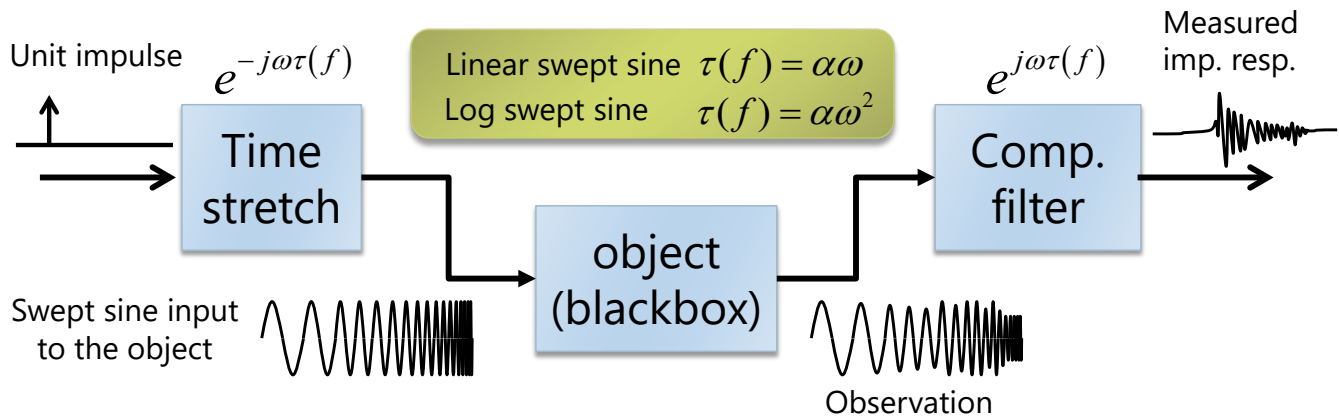


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Swept sine signal

- ▶ Swept sine is given by **stretching** a unit impulse, which is equivalent to applying **phase shift** to each frequency component of a unit impulse. The amount of phase shift is a function of the frequency.
- ▶ The effect of the *stretch* is compensated by applying a filter to the observation of the object.

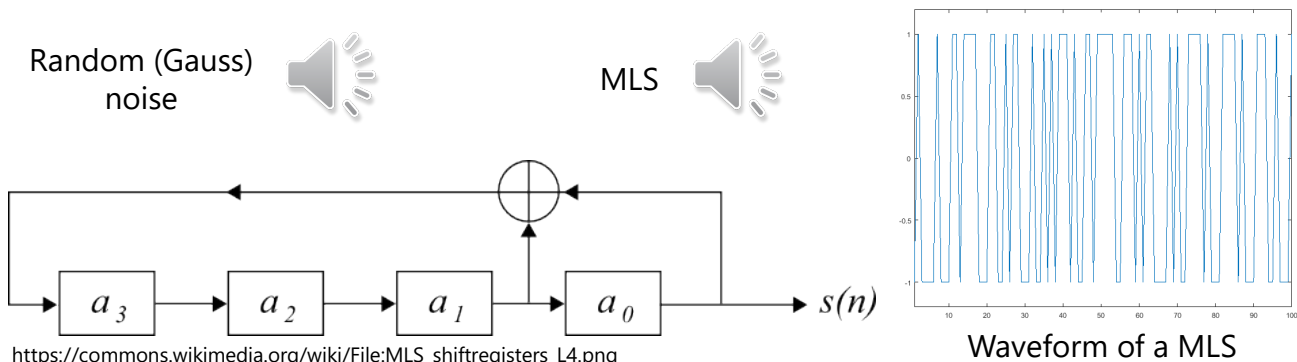


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Pseudo random noise

- ▶ Pseudo random noise is a binary sequence generated with deterministic algorithm that exhibits statistical behaviour similar to a truly random sequence.
- ▶ **Maximum length sequence (MLS)** is a type of pseudo random noise. As its power spectrum shows a quasi-uniform distribution, MLS is also used as the input for impulse response measurement.

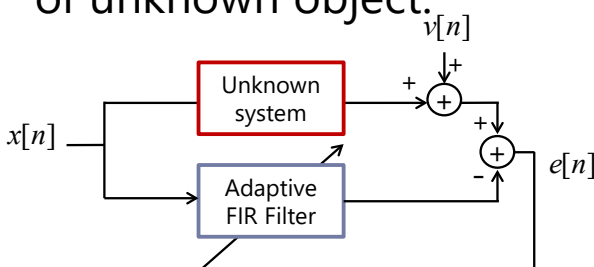


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What if you cannot choose the input?

- ▶ If the input to the object is already fixed, there are other ways to estimate the impulse response.
- ▶ **Adaptive filter** is a specific class of digital (FIR) filter that optimises its property to minimize a targeted value. It can be used to identify the impulse response of unknown object.
- ▶ **Cross-spectrum** measures the similarity between two signals. It can be used to estimate the frequency response of unknown object.



Cross-spectrum of input and output signals

$$H(\omega) = \frac{E[X(\omega)Y^*(\omega)]}{E[|X(\omega)|^2]}$$

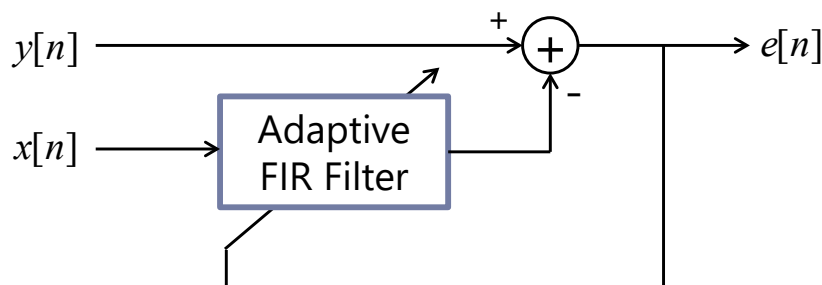


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Aside: Adaptive filter

- ▶ With adaptive filtering, filter coefficients of a digital filter is adjusted to optimise the performance of filter.
- ▶ Fundamental structure of adaptive filter is



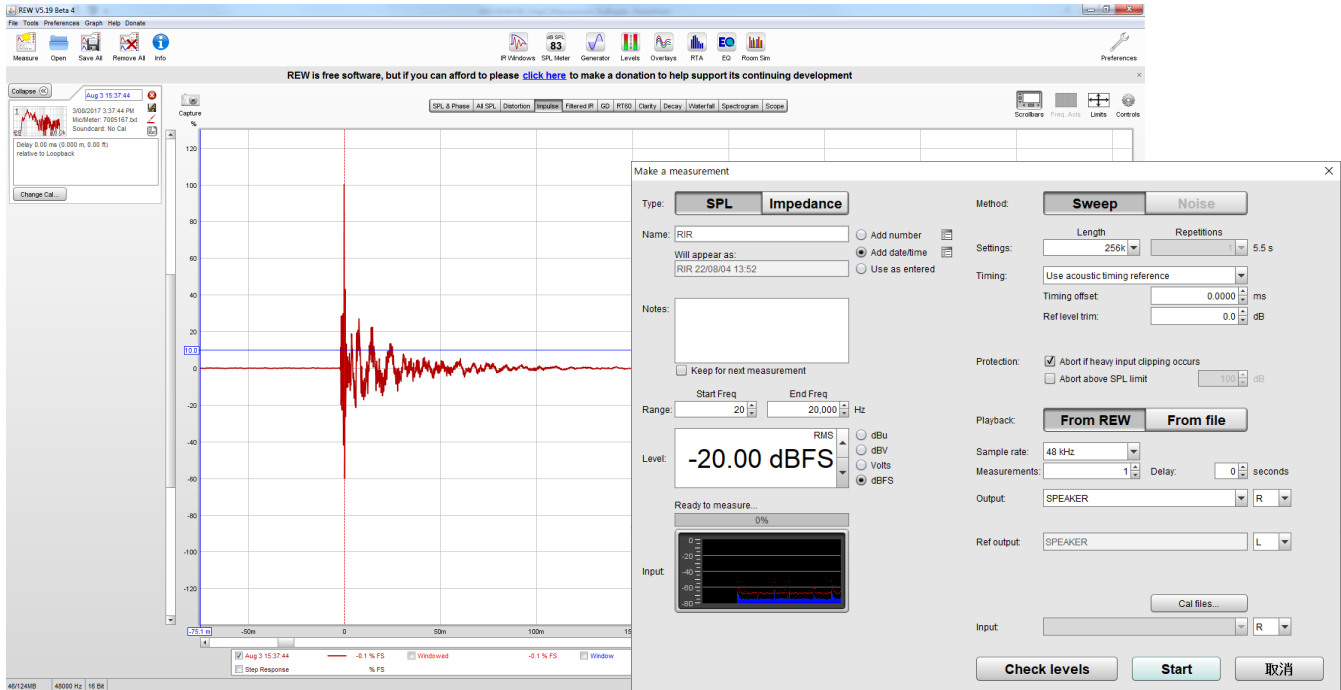
- ▶ Coefficients of the FIR filter is adjusted to minimise the error signal $e[n]$.

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Practical example

► Impulse response measurement using swept sine (REW)



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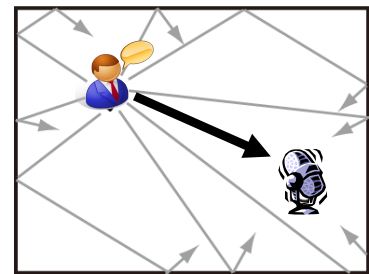
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Section 3: Room acoustics property measurement

Room Impulse Response Analysis

Analysing room impulse response

- ▶ As we have already seen, room impulse responses (RIR) describe the transmission channel between the source and observation point in a room.
- ▶ This implies that acoustical properties of the room can be investigated by analysing the measured RIR.
- ▶ Various metrics are available to quantify the properties which are calculated by processing the measured RIR.
- ▶ In this chapter we focus on the metrics that address:
 - ▶ Reverberation
 - ▶ Speech intelligibility
 - ▶ Binaurality

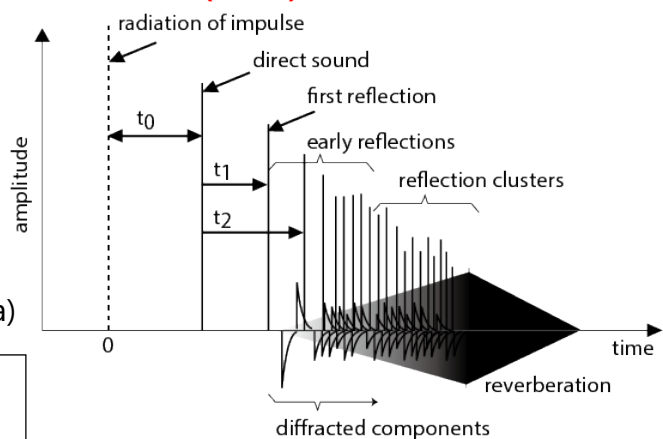


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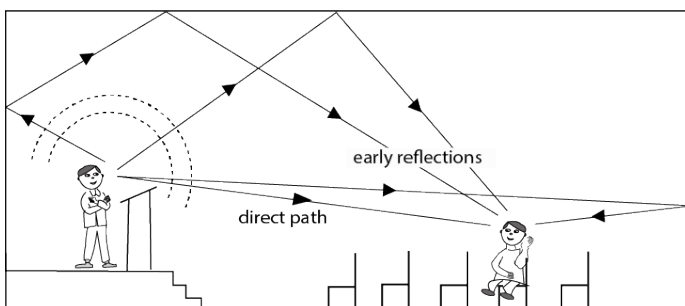
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RIR in a reverberant room

- ▶ A RIR measured in a reverberant room normally consists of **direct sound**, **early reflections**, and **(late) reverberation**.



Pulkki, Fig.2.20(a)



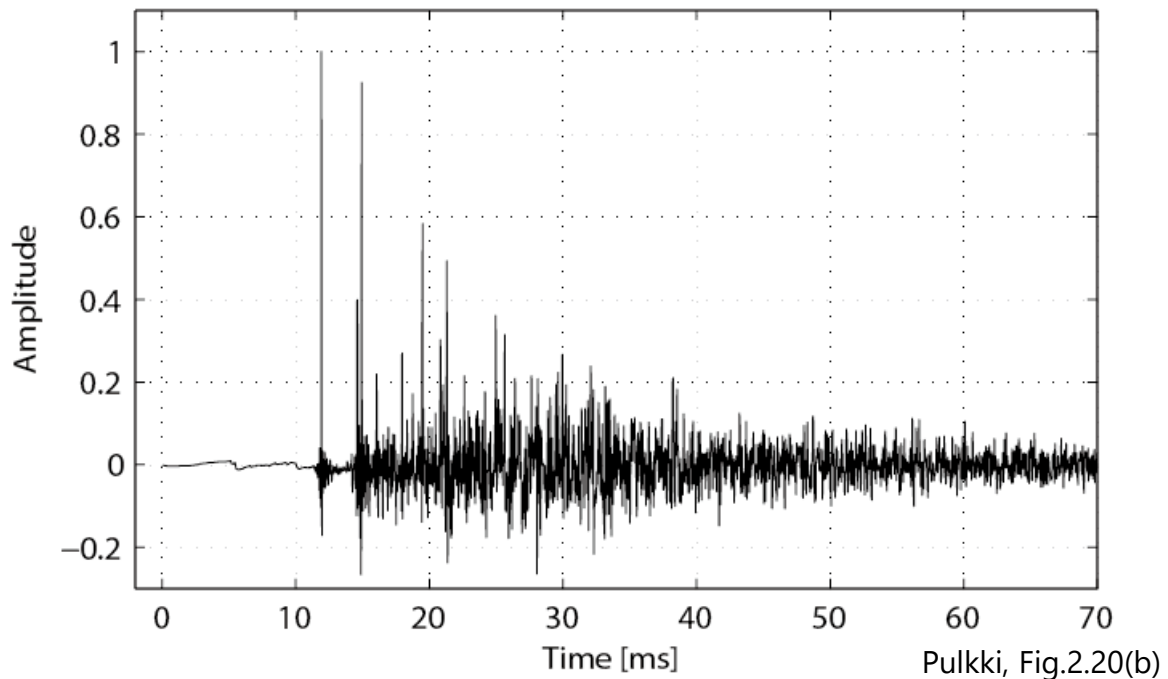
Pulkki, Fig.2.19

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RIR in a reverberant room

- ▶ A practically measured RIR looks like this.



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Question

- ▶ Question: Same speech was recorded in different rooms. Compared to Room A, to what extent was the room more/less reverberant?

Room A



Room B



How about this?

Room C

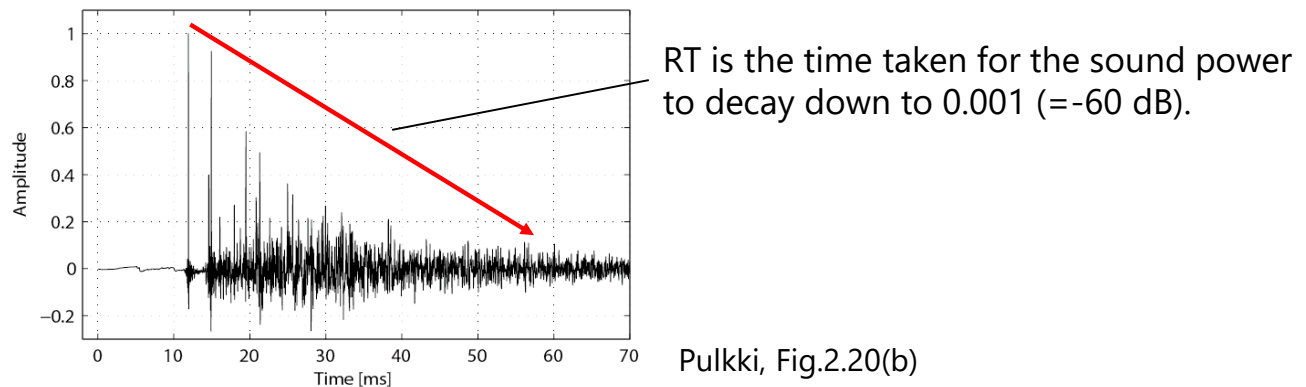


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Reverberation time

- ▶ Reverberation is one of the key properties that acoustics engineers are concerned about when considering acoustics of a room/auditorium.
- ▶ **Reverberation time (RT)**, the time that it takes sound to decay 60 dB after offset called T_{60} , is often used to measure the degree of reverberation in the environment.



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Reverberation time

- ▶ There are two ways to estimate the reverberation time.
- ▶ One method is estimation using **Sabine's formula**.

$$T_{60} = 0.161 \frac{V}{S}$$

Volume of room [m^3]: V

Absorption area of room [m^2]: $S = \sum_i a_i A_i$

Surface area [m^2]: A_i

Absorption coefficient: a_i

Note: the constant 0.161 has the units s/m.

▶ Example

A concert hall has a 5-meter tall ceiling, is 20 meters wide and 10 meters deep. The absorption coefficient for the walls, ceiling, and floors is assumed to be 0.3. Derive the RT using the Sabine's formula.

$$V = 10 \times 20 \times 5 = 1000 \text{ m}^3$$

$$S = 0.3 \times \{2 \times (10 \times 20) + 2 \times (10 \times 5) + 2 \times (5 \times 20)\} = 210 \text{ m}^2$$

$$T_{60} = 0.161 \frac{1000}{210} = 0.537 \text{ sec}$$

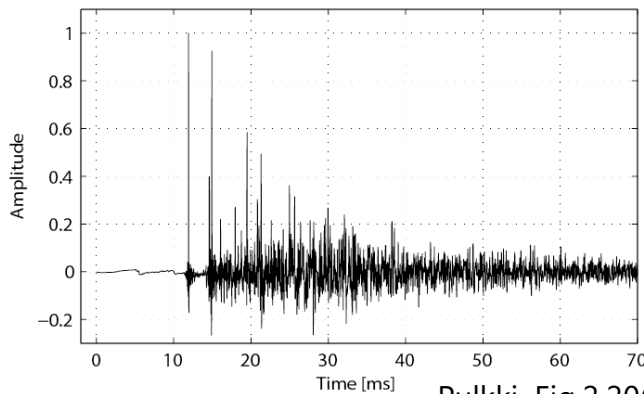
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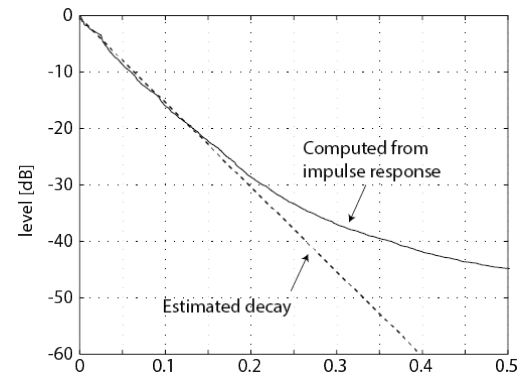
Reverberation time

- ▶ Sabine’s formula can only be used for *ideal* rooms.
- ▶ In practice, RT is often estimated from the measured RIR.
- ▶ **Early decay curve*** (EDC) is calculated from the RIR and its decay slope (T_{60}) is estimated by **linear regression**.

Early Decay Curve (EDC)
$$EDC(t) = 10 \log_{10} \left(\int_t^{\infty} h^2(\tau) d\tau / \int_0^{\infty} h^2(\tau) d\tau \right)$$



Pulkki, Fig.2.20(b)



Pulkki, Fig.2.21

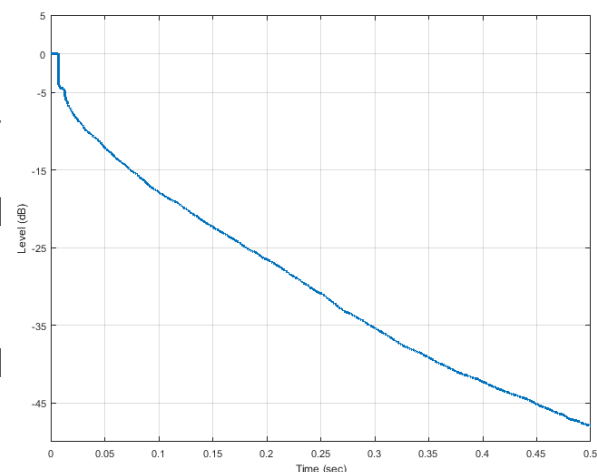
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Reverberation time

- ▶ In practice, the EDC often does not have sufficiently long straight part for estimating the T_{60} .
- ▶ A few different definitions of T_{60} are available depending on which part of EDC is used for the linear regression.
 - ▶ **Early decay time** (EDT) uses the section between 0 dB and -10 dB below the initial level of the EDC.
 - ▶ **T20** uses the section between -5 dB and -25 dB below the initial level of the EDC.
 - ▶ **T30** uses the section between -5 dB and -35 dB below the initial level of the EDC.

NB: T20 and T30 still indicate the time taken for the energy to decay 60 dB, but not 20 or 30 dB.



Practical early decay curve

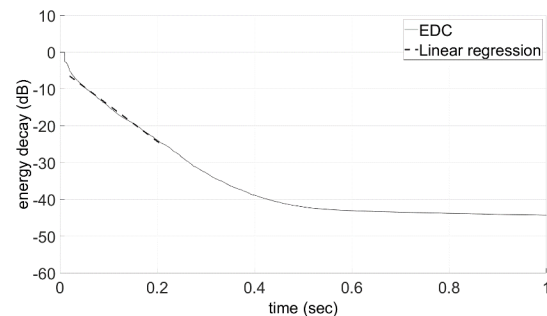
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Exercise

- ▶ When the result of linear regression applied to different regions of an EDC, calculate the reverberation time T_{60} of this room using the T20 measure.
 - ▶ Energy decay curve from 0 to -10 dB: $y = -150x - 1$
 - ▶ Energy decay curve from -5 to -25 dB: $y = -120x - 5$
 - ▶ Energy decay curve from -5 to -35 dB: $y = -90x - 5$

Note: x and y denote time (sec) and energy decay (dB).



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Reverberation time

- ▶ Recommendable T_{60} values:
 - ▶ Anechoic chamber: ~ 0.0 sec
 - ▶ Listening test room: 0.35 sec
 - ▶ Meeting room: 0.3 – 0.7 sec
 - ▶ Speech auditorium: 0.5 – 1.0 sec
 - ▶ Chamber music halls: ~ 1.4 sec
 - ▶ Large concert halls: ~ 2 sec
- ▶ RT does not vary for different positions in highly reverberant rooms whereas it changes a lot depending on position in rooms with absorbents.

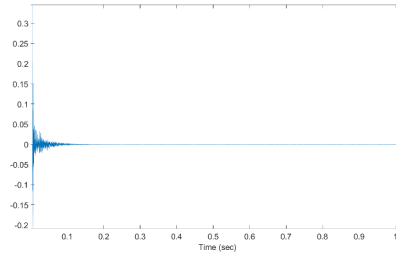
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Question

► Answer

Room A

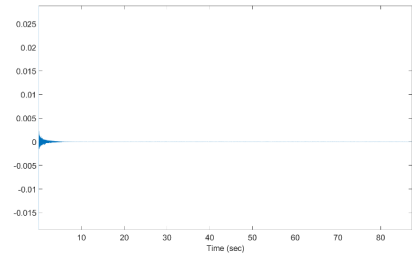


$T_{60}(T20): \sim 200\text{ms}$

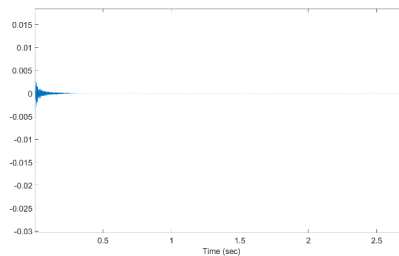
Room B



$T_{60}(T20): \sim 15\text{s}$



Room C



$T_{60}(T20): \sim 500\text{ms}$

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Another question

- Question: Are these recordings measured in the same room?

Recording 1



$T_{60}(T20): \sim 500\text{ms}$

Recording 2



$T_{60}(T20): \sim 500\text{ms}$

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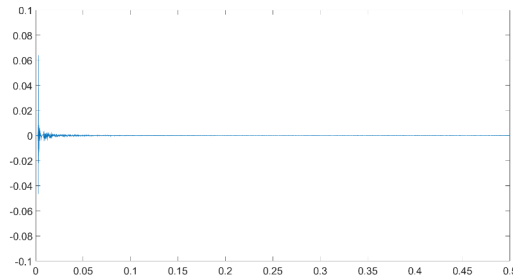
Another question

► Answer

Recording 1



$T_{60}(T20)$: ~500ms

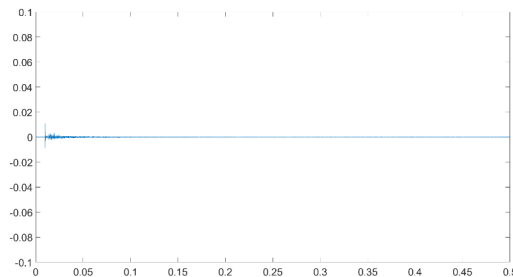


0.5 m from
loudspeaker

Recording 2



$T_{60}(T20)$: ~500ms



3.0 m from
loudspeaker

Yes they are, but recorded **at different distances** from the loudspeaker.

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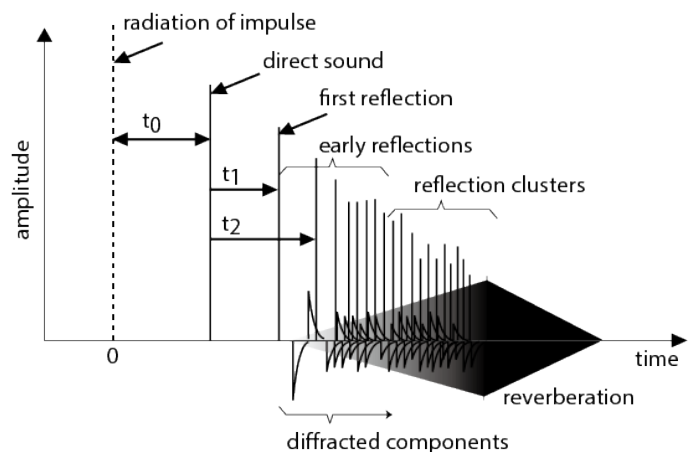
Direct-to-Reverberation ratio

- **Direct-to-reverberation-ratio (DRR)** is another metric associated with reverberation in a room.
- DRR is given by the ratio between the energy of direct sound and reverberation.

$$\text{DRR} = \frac{\text{Energy of direct sound}}{\text{Energy of reverberant sound}}$$

$$= 10 \log_{10} \left(\frac{\int_t h_D^2(t) dt}{\int_t h_R^2(t) dt} \right) \text{ dB}$$

$$= 10 \log_{10} \left(\frac{\int_0^\tau h^2(t) dt}{\int_\tau^\infty h^2(t) dt} \right) \text{ dB}$$



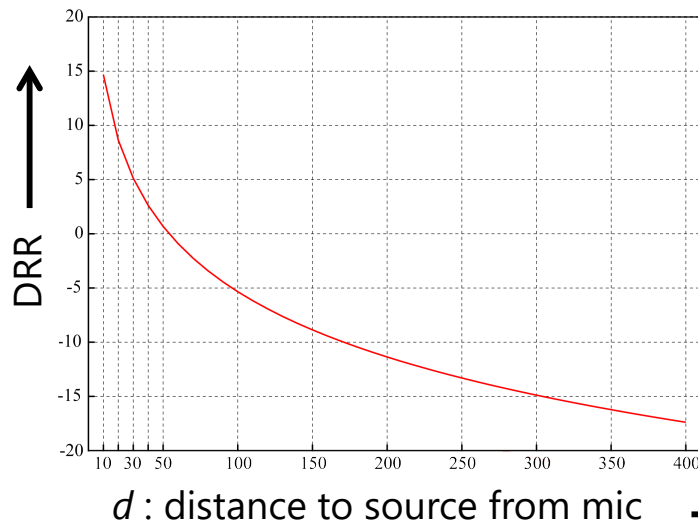
Pulkki, Fig.2.20(a)

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Direct-to-Reverberation ratio

- ▶ It is known that human hearing is estimating **distance** of sound source using DRR.
- ▶ There is one-to-one relationship between DRR and source distance.



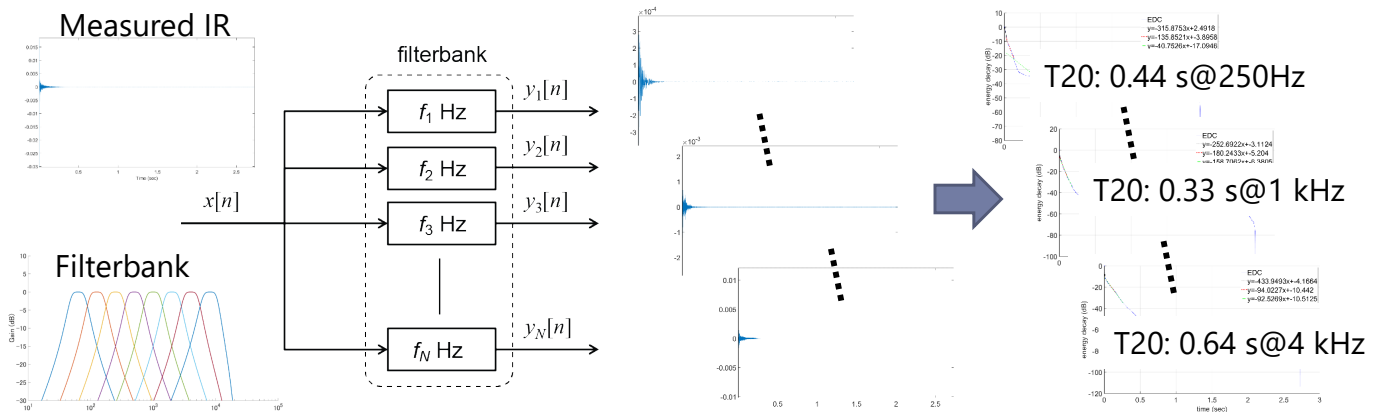
example:
DRR curve in a room
(4×6×2.5 [m], $\alpha = 0.15$)

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Octave-band analysis

- ▶ Metrics derived from measured impulse response can also be calculated for each octave (or 1/3 octave band).
- ▶ This is realised by applying octave or (1/3 octave band) **filterbank** to the measured impulse response before implementing the metric calculation.
- ▶ Example: Octave-band reverberation time



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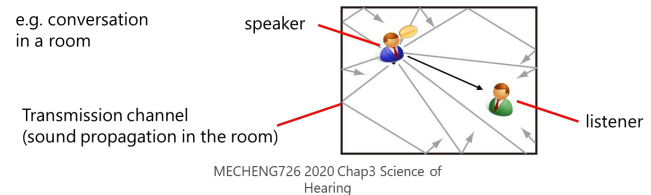
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Speech intelligibility measurement

- ▶ High intelligibility is of crucial importance in telephones, public announcement systems, and in auditoria.
- ▶ As an acoustic engineer, **quantifying** speech intelligibility of environment is often essential for better acoustical design.
- ▶ Technical interpretation of intelligibility is related to the transmission channel.

Speech intelligibility

- ▶ **Speech intelligibility** (or simply intelligibility) is a property referring to how well the meaning of a spoken message is transmitted to a listener.
- ▶ Intelligibility depends on:
 - ▶ Ability of the **speaker** to produce acoustically and linguistically clear message.
 - ▶ How well the **transmission channel** is able to transmit the message.
 - ▶ How well the **listener** is able to receive and analyse the message.



Speech intelligibility measurement

- ▶ There are two ways to evaluate intelligibility.
- ▶ Subjective methods (i.e. listening by humans)
 - ▶ Human participants evaluate sounds through rating/cognitive tasks.
 - ▶ Reflects actual comprehension of speech by humans.
 - ▶ Arranging sufficient number of participants is a tedious task.
 - ▶ No repeatability. May be affected by individual participant's ability.
- ▶ Objective methods (i.e. measuring quantitative metrics)
 - ▶ Use quantitative metrics.
 - ▶ Good repeatability. Easy to evaluate.
 - ▶ Results may not reflect actual human comprehension.

Speech intelligibility

- ▶ **Speech transmission index (STI)** is an objective metric that estimates the intelligibility of a transmission channel.
- ▶ STI is based on the **modulation transfer function (MTF)**, which can be calculated from the impulse response of the transmission channel (i.e. system).
- ▶ Process for STI measurement are standardised as IEC 60268-16.
 - ▶ Calculate MTF

$$\text{MTF} \quad m_k(\omega) = \frac{\int_0^\infty h_k^2(t) e^{-j\omega t} dt}{\int_0^\infty h_k^2(t) dt} \quad h_k(t): \text{impulse response filtered by } k\text{-th octave band BPF}$$

- ω is the 14 centre frequencies of the 1/3 octave band between 63 Hz and 12.5 kHz.
- k is 7 octave bands between 125 Hz and 8 kHz.

Speech intelligibility

- ▶ MTF is converted to the apparent signal-to-noise ratio (SNR).

$$\text{Apparent SNR} \quad \text{SNR}_k(\omega) = 10 \log_{10} \frac{m_k(\omega)}{1 - m_k(\omega)}$$

- ▶ MTF is converted to another metric called transmission index (TI).

Transmission Index

$$K_k(\omega) = \begin{cases} 1 & 15 < \text{SNR}_k(\omega) \\ (\text{SNR}_k(\omega) + 15) / 30 & -15 \leq \text{SNR}_k(\omega) \leq 15 \\ 0 & \text{SNR}_k(\omega) < -15 \end{cases}$$

Speech intelligibility

- ▶ TI's calculated at different ω are averaged providing the following value M_k .
(Remember ω is the 14 centre frequencies of the 1/3 octave band between 63 Hz and 12.5 kHz.)

$$M_k = \frac{1}{14} \sum_{\omega} K_k(\omega)$$

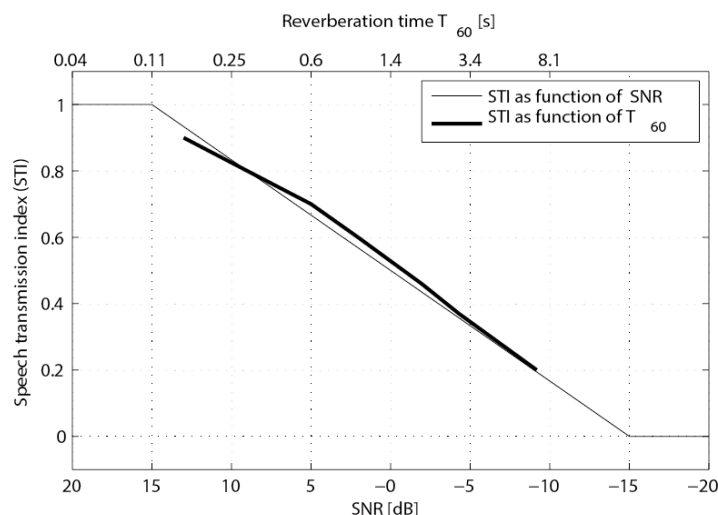
- ▶ Finally STI is calculated by weighted average of M_k across octave bands (Remember 7 octave bands between 125 Hz and 8 kHz.)

$$STI = \sum_k W_k M_k$$

	W_1	W_2	W_3	W_4	W_5	W_6	W_7
Weight	0.129	0.143	0.114	0.114	0.186	0.171	0.143
Octave band (Hz)	125	250	500	1000	2000	4000	8000

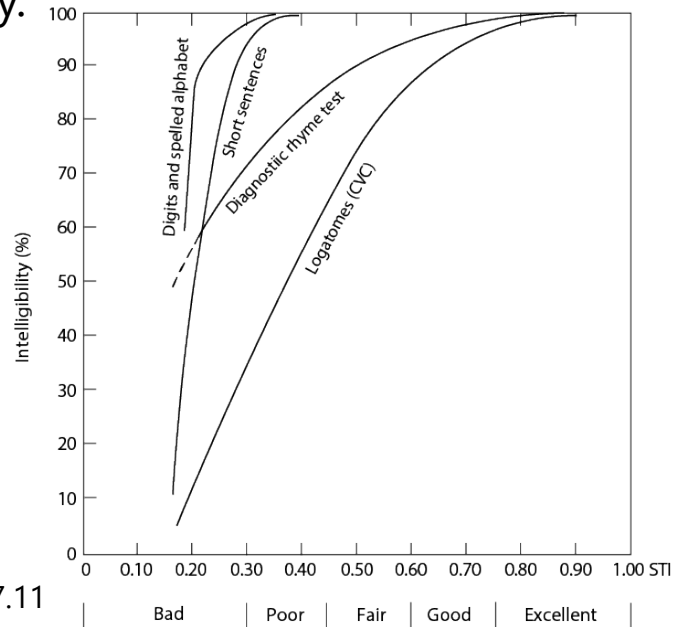
Speech intelligibility

- ▶ Speech intelligibility is affected by **reverberation** of the room.
- ▶ The effect of the apparent SNR on the STI are similar to the effect of reverberation (T_{60}) on the STI.



Speech intelligibility

- ▶ Speech intelligibility is also affected by the **type of speech**.
- ▶ This implies linguistic ability of individuals would also affect the speech intelligibility.



Pulkki, Fig.17.11

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Other metrics related to intelligibility

- ▶ Apart from STI, there are several other objective metrics calculated from the impulse response of the channel that are related to intelligibility.
 - ▶ **Clarity** measures the energy ratio between the early and late responses, i.e. same as DRR, but with specific values used for τ . It represents perceived clarity of sound in an auditorium/room.
 - ▶ C_{50} with $\tau = 50$ ms is more suitable for evaluating **speech** intelligibility whereas C_{80} with $\tau = 80$ ms is more suitable for evaluating **music** intelligibility.

$$C_{50} = 10 \log_{10} \frac{\int_0^{50ms} h^2(t) dt}{\int_{50ms}^{\infty} h^2(t) dt}$$

$$C_{80} = 10 \log_{10} \frac{\int_0^{80ms} h^2(t) dt}{\int_{80ms}^{\infty} h^2(t) dt}$$

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Other metrics related to intelligibility

- ▶ **Definition** measures the energy ratio for the first 50 ms of the impulse response to the entire impulse response.

$$D_{50}(\%) = \frac{\int_0^{50ms} h^2(t) dt}{\int_0^{\infty} h^2(t) dt}$$

- ▶ Definition can also be derived from Clarity C_{50} .

$$C_{50}(dB) = 10 \log_{10} \frac{D_{50}}{1 - D_{50}}$$

Binaurality

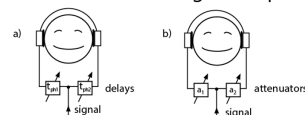
- ▶ Room acoustics properties that affect spatial hearing are also acquired from the measured impulse responses.
- ▶ **Binaurality** (condition for binaural hearing) is realised when the room impulse responses measured at the positions of ears carry binaural cues such as ITD and ILD.

Spatial hearing

- ▶ **Spatial hearing** refers to human's ability to **localise** sound source using our (two) ears.
- ▶ Localisation can be described by the perception of the source's **direction**, **distance**, and **spatial extent**.
- ▶ Two concepts used in spatial hearing.
 - ▶ **Monastral hearing**: listening in conditions where there is no interaural difference information, e.g. same signal enters both ears.
 - ▶ **Binaural hearing**: listening to sounds where information due to interaural differences exists and is taken into account.
- ▶ More information about the acoustical environment (acoustic cues) is provided in binaural hearing.

Binaural hearing

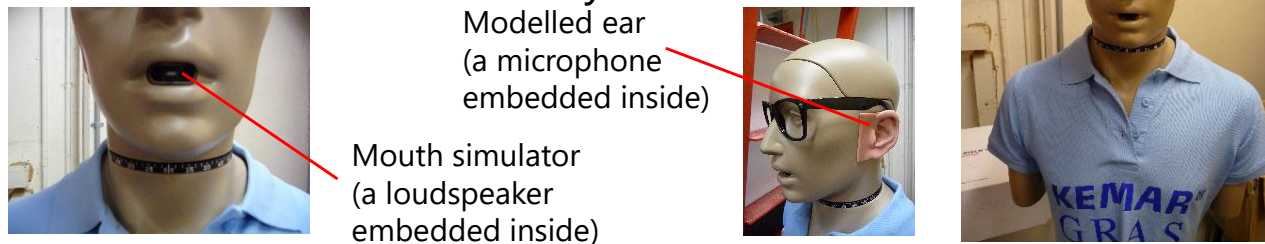
- ▶ Binaural cues are derived from the difference in the signals between the two ears. There are two main acoustic cues:
 - ▶ **Interaural time difference (ITD)** occurs due to difference of distance for a sound to propagate from the source to the ears.
 - ▶ **Interaural level difference (ILD)** occurs due to the scattering (shadowing, reflection) of sound waves by the head and torso.
- ▶ Binaural hearing can be offered by projecting sound with controlled ITD/ILD through headphones.



Pulkki, Fig.12.1

Binaurality

- ▶ **Head related impulse response (HRIR)** represents the effects of the torso, head, and the external ear acoustics.
- ▶ **Head related transfer function (HRTF)** is the frequency response derived from the HRIR.
- ▶ HRIR/HRTF is measured using a dummy head that replicates the average shape of human's head and torso.
- ▶ If the HRIR is available, binaural hearing can be easily reproduced by convolving the HRIR with an arbitrary sound source.

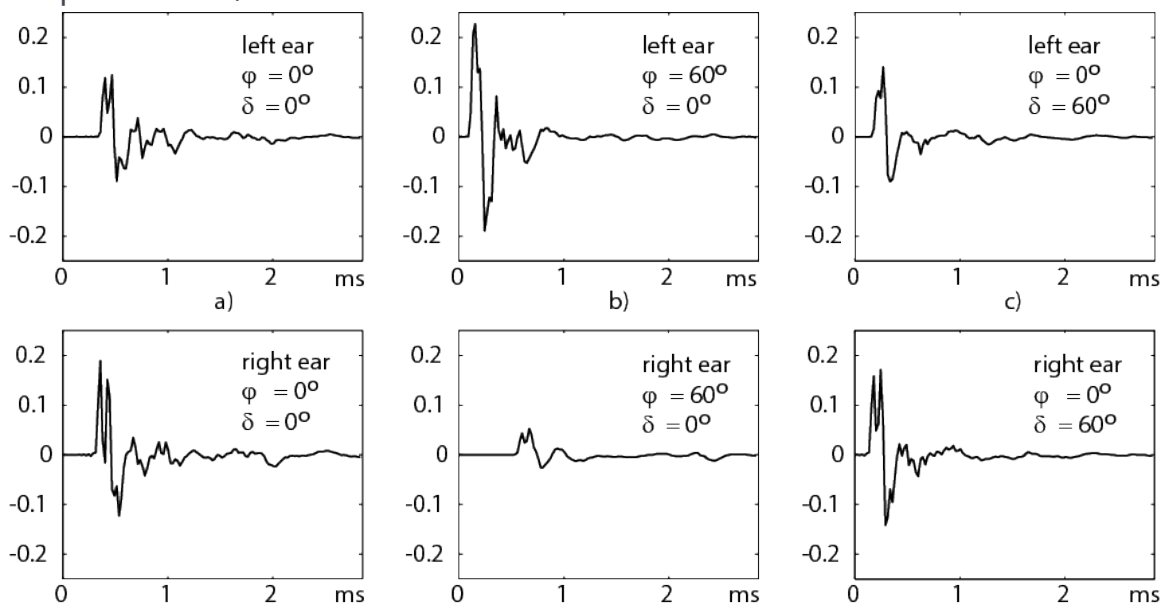


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Binaurality

- ▶ A set of HRIRs measured from different angles
 - ▶ φ : azimuth, δ : elevation



Pulkki, Fig.12.5

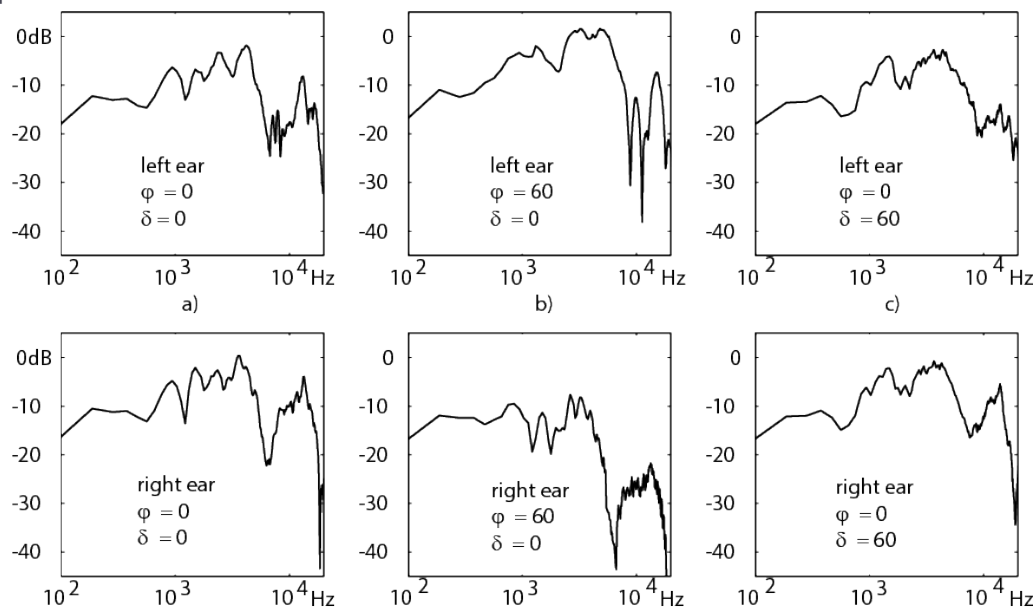
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Binaurality

- ▶ A set of HRTFs measured from different angle

- ▶ φ : azimuth, δ : elevation



Pulkki, Fig.12.6

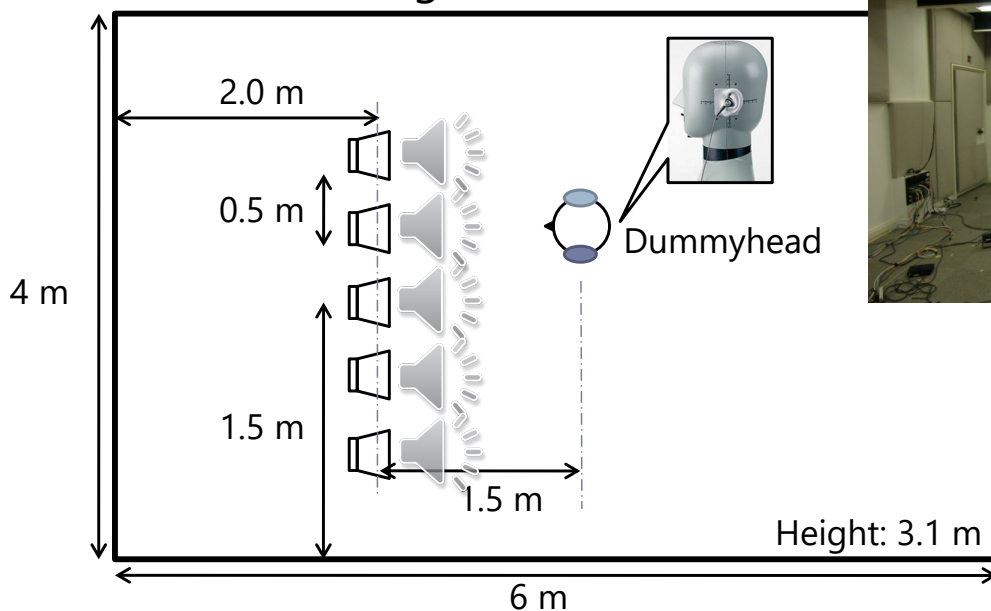
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Binaurality

- ▶ Example of binaural recording

Listen to the files using headphones. Does the speaker localise at the right direction?



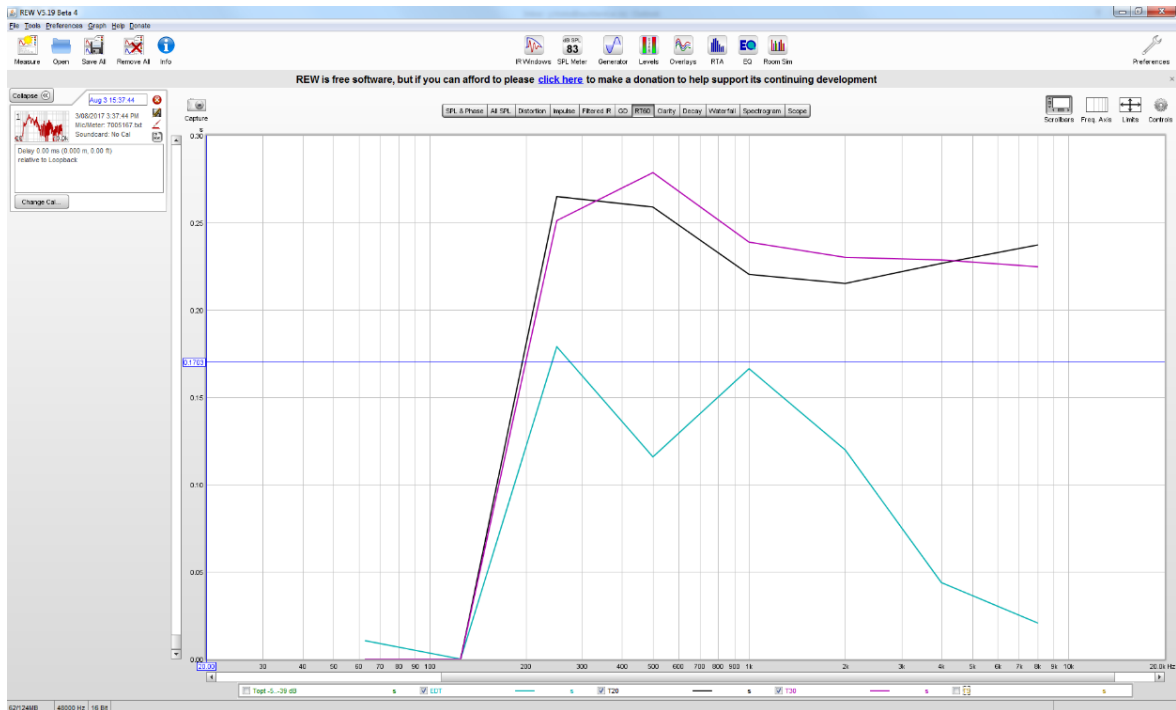
$T_{60} \approx 250-300\text{ms}$

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Practical example

► Estimating T_{60} (REW)



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Practical example

► Stereo recording



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